

AZURE DATA ENGINEERING

ZERO TO HERO — SYLLABUS ROADMAP

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Fundamentals → SQL → Azure → Storage → ADF → Synapse → Python → Spark/PySpark → Databricks → Delta Lake → Fabric → AI

1. DATA ENGINEERING FUNDAMENTALS

- What is Data Engineering
- Introduction to Azure
- Azure vs AWS Data Engineering services
- Big Data introduction

2. SQL

- SQL fundamentals
- Database, table, row and column concepts
- RDBMS basics
- Primary Key and Foreign Key
- DDL, DML, DCL and TCL
- CREATE, ALTER, DROP and TRUNCATE
- Constraints: PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, DEFAULT
- SELECT, INSERT, UPDATE and DELETE
- WHERE, AND, OR, NOT, IN, BETWEEN and LIKE
- GRANT, REVOKE, user roles and permissions
- COMMIT, ROLLBACK, SAVEPOINT and ACID properties
- Aggregate, scalar, string, date and numeric functions
- Joins: INNER, LEFT, RIGHT, FULL, CROSS and SELF JOIN
- UNION, UNION ALL, INTERSECT and EXCEPT
- GROUP BY, HAVING, ORDER BY and DISTINCT
- Subqueries and CTEs
- Views and materialized views
- Stored procedures and parameters
- User-defined functions
- Indexes and performance
- Query optimization and execution plans
- Window functions: ROW_NUMBER, RANK, DENSE_RANK, LEAD and LAG
- CASE statements
- PIVOT and UNPIVOT
- Temporary tables vs table variables

3. AZURE

- What is Azure and Cloud
- Resource Groups
- Azure services overview
- Azure Portal walkthrough

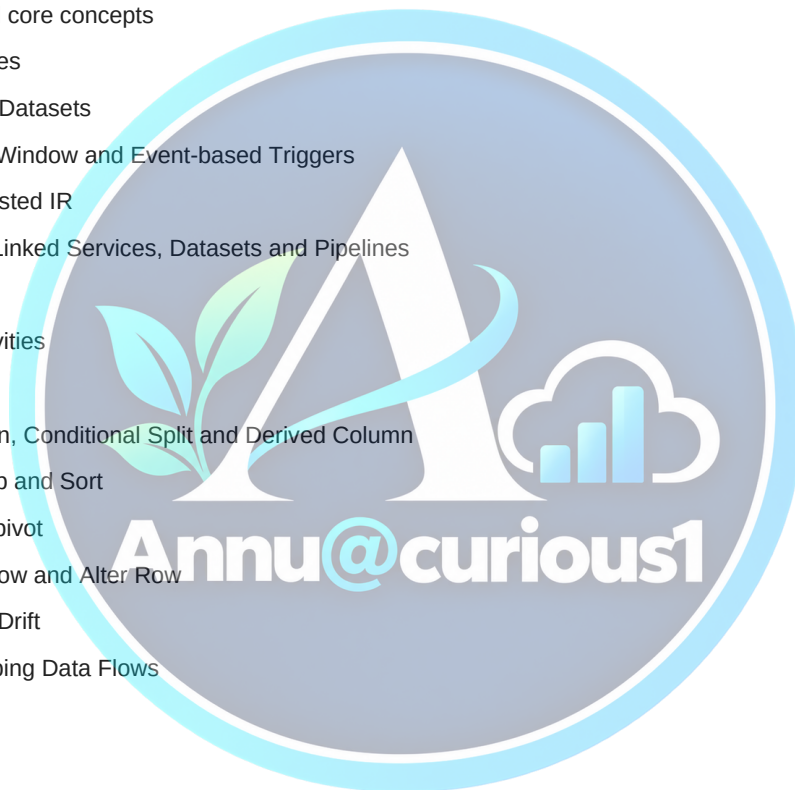
- SDKs and tools for Azure resources
- Azure subscription basics

4. AZURE STORAGE & DATA LAKE

- Azure Blob Storage
- Containers and blobs
- Upload, list, download and delete blobs
- Azure Data Lake Storage Gen2
- ADLS Gen2 accounts, folders and data upload
- Data security
- Azure Key Vault
- Storing secrets using Azure Key Vault

5. AZURE DATA FACTORY

- ADF introduction and core concepts
- Pipelines and Activities
- Linked Services and Datasets
- Schedule, Tumbling Window and Event-based Triggers
- Azure IR and Self-hosted IR
- Parameterization of Linked Services, Datasets and Pipelines
- System Variables
- Connectors and Activities
- Mapping Data Flow
- Filter, Aggregate, Join, Conditional Split and Derived Column
- Exists, Union, Lookup and Sort
- Select, Pivot and Unpivot
- Surrogate Key, Window and Alter Row
- Flatten and Schema Drift
- Parameterizing Mapping Data Flows
- Validate Schema
- Wrangling Data Flow
- GitHub and Azure DevOps integration
- Continuous Integration and Deployment (CI/CD)
- Azure Key Vault integration
- Passing activity output between activities
- Annotations and Global Parameters
- Fail Activity, Inline Dataset and Assert
- Flowlets and Script Activity
- Fuzzy Joins and Cast transformation
- Web/table extraction
- ADF billing and performance concepts
- Alerts for pipeline/activity failures
- Pipeline return values

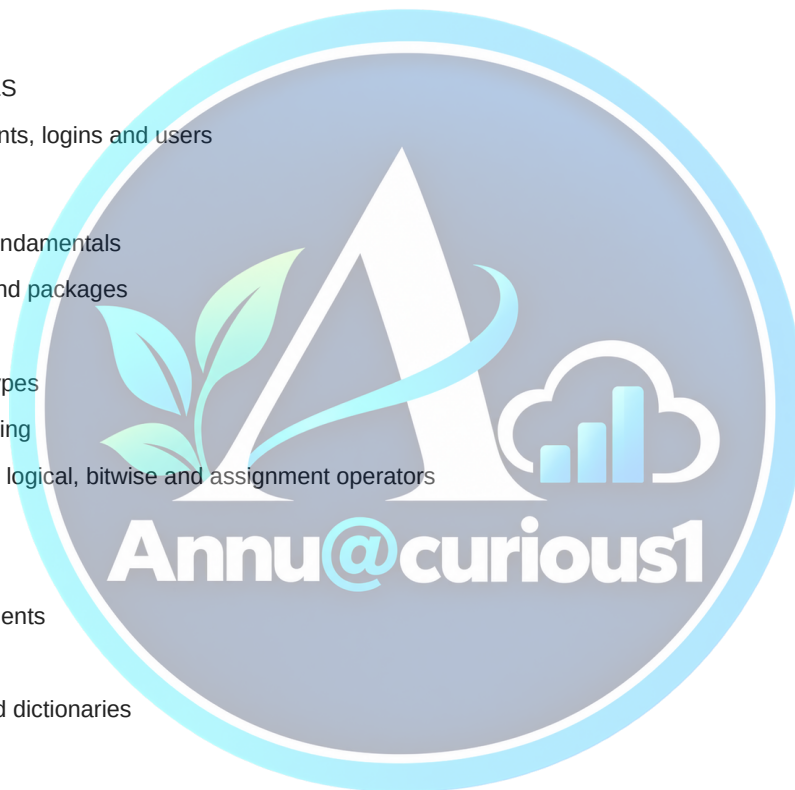


6. AZURE SYNAPSE

- Synapse Analytics introduction
- Creating a Synapse Workspace
- Synapse fundamentals
- Serverless SQL Pool and Dedicated SQL Pool
- Serverless Spark Pool
- Analyzing data in ADLS through Synapse
- Synapse Pipelines
- Workspace monitoring
- Synapse SQL architecture
- Distributions: Hash, Round Robin and Replicate
- External Data Sources and External File Formats
- CETAS and CTAS
- External Tables
- Querying files in ADLS
- Administrative accounts, logins and users

7. PYTHON

- Python syntax and fundamentals
- Development tools and packages
- Environment setup
- Variables and data types
- Strings and type casting
- Arithmetic, relational, logical, bitwise and assignment operators
- if / elif / else
- while and for loops
- Functions and arguments
- Lambda functions
- Lists, tuples, sets and dictionaries
- File I/O
- Reading and writing files



8. SPARK / PYSPARK

- Spark overview, features and use cases
- Spark vs Hadoop MapReduce
- Spark Core, Spark SQL and Spark Streaming
- Spark architecture and application flow
- Driver, Executors and Cluster Managers
- Execution modes
- RDDs and RDD features
- Transformations and Actions
- RDD partitions and coalesce
- Cache and persist

- Aggregations and joins
- Broadcast variables and accumulators
- Spark Structured APIs
- DataFrames, schemas, columns and rows
- DataFrame operations, aggregations and joins
- CSV, JSON, Parquet, ORC, text and SQL data sources
- Spark SQL and Hive integration
- PySpark development tools
- Batch vs Streaming
- Structured Streaming
- Event Time and Stateful Processing

9. AZURE DATABRICKS

- Azure Databricks introduction
- Creating a Databricks Workspace
- Workspace assets
- Spark Jobs in Databricks
- Databricks architecture
- DBFS and Databricks Utilities
- Notebook utilities and widgets
- Notebook parameters
- Mount points
- Secret Scopes
- Databricks CLI basics
- Azure Key Vault-backed secret scopes
- Accessing ADLS/Blob using account keys, service principals and SAS

10. DELTA LAKE

- Delta Lake architecture
- Storage concepts
- Delta table creation and APIs
- DML operations
- Partitioning
- Schema Enforcement
- Schema Evolution
- Versions and Time Travel
- VACUUM
- Delta MERGE
- SCD Type 1 and SCD Type 2
- On-premises SQL Server to Lakehouse ingestion
- Incremental ingestion
- Logging pipeline activity
- End-to-End ADF pipelines and Dataflows Gen2



11. MICROSOFT FABRIC

- Microsoft Fabric introduction
- Fabric Workspace
- Lakehouse
- OneLake Explorer and Data Studio
- Fabric Warehouse
- Lakehouse vs Warehouse
- Fabric Data Factory experience
- Pipelines and Dataflows
- Apache Spark in Fabric
- Delta Lake tables in Fabric
- REST API to Lakehouse using Bearer Token
- Moving folders from ADLS Gen2 to Lakehouse

12. AI

- Introduction to Data Agents
- AI-powered assistants for data interaction
- Natural-language interaction with enterprise data
- Microsoft Copilot and Azure OpenAI in Microsoft Fabric
- Traditional BI vs AI Data Agents
- Natural-language querying
- Data discovery
- Automated insights
- Conversational analytics
- Business-data semantic understanding
- Databricks Genie
- Natural-language-to-SQL generation
- LLM-based insight generation
- Unity Catalog metadata integration

